

Auditing the Wind Tunnel: the Twin–Enrichment Closed Form at Reachable ω , and the Marginal–versus–Per-Centre Distinction

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Code, data, and figures: github.com/Ruqing1963/6N-windtunnel-audit

Abstract

Part XXIII extrapolated the twin–centre enrichment multiplier $\prod_{q|N} q/(q-2)$ to the primordial depth $\omega = 15$, obtaining $\times 7.03$, and flagged the step honestly as a closed-form self-portrait assuming local-factor multiplicativity persists past the $\omega \approx 5$ – 6 range where it had been checked. We audit that assumption directly. Sieving all centres to $6N \leq 2 \times 10^8$ (813,370 twin centres) we measure the actual enrichment, factor by factor. Three findings. (i) The microscopic *marginal* law is exact: the twin rate among centres divisible by q exceeds the global rate by $q/(q-2)$, verified to $\lesssim 0.3\%$ for every $q = 5, \dots, 59$; the divisor/non-divisor ratio is the complementary $(q-1)/(q-3)$, reconciling the two forms used in Part I and Part XXIII as two views of one mechanism. (ii) Marginal multiplicativity holds within Poisson error for every prime combination with an adequate sample ($\lesssim 1\%$ through triples), with a faint systematic lean to the suppression side—the sign of the Part IV second-order residual, but within our statistics. (iii) The wind-tunnel object $\prod_{q|N} q/(q-2)$, however, is a *marginal* factor and is not the per-centre likelihood. The correct per-centre enrichment of a centre with divisor set D is $C \prod_{q \in D} (q-1)/(q-3)$ with a global constant $C = \prod_{q>3} \frac{q(q-3)}{(q-1)(q-2)} = 0.7216$; this matches the data to $< 1\%$ across the whole reachable range, whereas the wind-tunnel object over-predicts the per-centre rate by up to $\sim 38\%$ for typical factorisations. For the squarefree primordial centre the two nearly coincide (6.98 vs. 7.03, a 0.66% gap), so the headline $\times 7.03$ is sound *for the primordial centre and for the marginal divisibility statement*—but it is an upper envelope, not the typical ω -stratified enrichment, which sits far below it ($\langle E \rangle \approx 2.6$ at $\omega = 6$ versus the 4.27 envelope). The wind-tunnel mechanism is thus vindicated microscopically; the two corrections are interpretive (marginal vs. per-centre) and scope (envelope vs. typical). As throughout, this is a measurement, not a theorem, and no claim is made about the infinitude of any constellation.

1 Introduction

In Volume I the “wind tunnel” (Review §9.1) and the kinetics of Part XXIII synthesised centres with prescribed prime factorisation and applied the per-factor twin modifier $q/(q-2)$, taking the product over the ω prescribed factors to reach an enrichment of $\times 7.03$ at the $\omega = 15$ primordial ($N > 4.7 \times 10^{18}$) [7, 9]. Both papers labelled the figure plainly as a closed-form extrapolation rather than a measurement, and named the load-bearing assumption: that the local-factor multiplicativity, validated only to $\omega \approx 5$ – 6 , continues to hold. That is exactly the kind of past extrapolation this programme’s honest ledger asks us to return and test against data. This paper is that audit.

We do not reach $\omega = 15$ —no sieve does. We reach the ω that direct enumeration to $6N \leq 2 \times 10^8$ makes statistically solid ($\omega \lesssim 6$), and we test the *ingredients* of the extrapolation there: the per-prime modifier, its multiplicativity, and—the point that turns out to matter most—the precise sense in which $\prod q/(q-2)$ is and is not the enrichment of a given centre.

2 Method and two notions of enrichment

For a centre N let $\omega_{>3}(N)$ count the distinct prime factors > 3 of N , and call N a *twin centre* if $6N-1$ and $6N+1$ are both prime. We sieve all centres $N \leq 3.33 \times 10^7$ ($6N \leq 2 \times 10^8$), computing $\omega_{>3}$ by a distinct-prime sieve and twin membership from a prime sieve to 2×10^8 ; there are 813,370 twin centres, a global rate $p_0 = 0.0244011$.

For a prime $q > 3$, a twin centre is forbidden on the two killer residues $\text{dead}(q) = \{6^{-1}, -6^{-1}\} \pmod{q}$; by CRT the survival events at distinct primes are independent, and the all-centre survival factor at q is $(q-2)/q$. Two distinct conditioning questions then arise, and conflating them is the whole subject of this paper.

Definition 1 (Marginal enrichment). The ratio of the twin rate among centres with $q \mid N$ (all other primes unconstrained) to the global rate p_0 . Conditioning on a divisor *set* D marginalises every prime outside D .

Definition 2 (Per-centre enrichment). The ratio $\mathbb{E}[\text{twin} \mid F]/p_0$ where F is the *complete* factorisation of N —so that for *every* prime it is known whether it divides N . Binning by $\omega_{>3}(N)$ or by any function of F measures this notion.

3 The microscopic law is exact

Conditioning on $q \mid N$ fixes $N \equiv 0$, a safe residue, so survival at q becomes certain; relative to the baseline $(q-2)/q$ the factor is $q/(q-2)$. Conditioning instead on $q \nmid N$ leaves N uniform on the $q-1$ nonzero residues, of which 2 are killers, giving survival $(q-3)/(q-1)$ and a divisor/non-divisor ratio $(q-1)/(q-3)$. Table 1 confirms both to $\lesssim 0.3\%$ across $q = 5, \dots, 59$ (Fig. 1A). The $q/(q-2)$ form is the wind-tunnel modifier of Part XXIII; the $(q-1)/(q-3)$ form is the conditional enrichment tabulated in Part I [3, 7]. They are not in tension: they are the baseline-relative and the divisor-relative views of the same two-killer geometry, related identically by $\frac{q/(q-2)}{(q-1)/(q-3)} = g(q)$ with g as in §5.

4 Multiplicativity holds within statistics

The extrapolation’s core assumption is that distinct primes’ modifiers multiply. Under CRT independence this is exact for the marginal enrichment: conditioning on a divisor set D gives $\prod_{q \in D} q/(q-2)$. Table 2 tests it. Every combination with an adequate sample agrees with the product to $\lesssim 1\%$; the apparent -8.5% at $\{5, 7, 11, 13\}$ rests on only ~ 500 twins (relative Poisson error $\approx 4.5\%$, so $\sim 2\sigma$). The residuals carry a faint systematic lean to the suppression side, the sign of the second-order inter-centre residual isolated empirically in Part IV [4], but they do not exceed $\sim 2\sigma$ anywhere; within the reachable range, multiplicativity is confirmed and the wind-tunnel product is the right marginal object.

q	meas. $q/(q-2)$	pred. $q/(q-2)$	meas. $(q-1)/(q-3)$	pred. $(q-1)/(q-3)$
5	1.66427	1.66667	1.99568	2.00000
7	1.39738	1.40000	1.49650	1.50000
11	1.22072	1.22222	1.24827	1.25000
13	1.18470	1.18182	1.20322	1.20000
17	1.13888	1.13333	1.14885	1.14286
19	1.11622	1.11765	1.12347	1.12500
23	1.09346	1.09524	1.09812	1.10000
29	1.07675	1.07407	1.07971	1.07692
37	1.05909	1.05714	1.06083	1.05882
47	1.04376	1.04444	1.04475	1.04545
59	1.04019	1.03509	1.04091	1.03571

Table 1: Per-prime marginal enrichment at $6N \leq 2 \times 10^8$. Both closed forms hold to $\lesssim 0.3\%$; deviations change sign with q and are consistent with statistical noise.

divisor set D	n centres	meas. enrich	$\prod_{q \in D} \frac{q}{q-2}$	resid. (%)
$\{5, 7\}$	952380	2.3256	2.3333	-0.33
$\{5, 11\}$	606060	2.0272	2.0370	-0.48
$\{7, 11\}$	432900	1.6944	1.7111	-0.98
$\{5, 13\}$	512820	1.9712	1.9697	+0.07
$\{5, 7, 11\}$	86580	2.8424	2.8519	-0.33
$\{5, 7, 13\}$	73260	2.7109	2.7576	-1.69
$\{5, 7, 11, 13\}$	6660	3.0829	3.3704	-8.53

Table 2: Marginal multiplicativity. Agreement is $\lesssim 1\%$ wherever the twin sample exceeds $\sim 10^4$; the last row is sample-starved (~ 500 twins).

5 The marginal-per-centre distinction (the substantive correction)

The wind tunnel reports $\prod_{q|N} q/(q-2)$ as “the enrichment of a centre” with that factorisation. But $q/(q-2)$ is a *marginal* factor, and a per-centre likelihood must also account for the primes that do *not* divide N : knowing the complete factorisation F says that each $q \nmid N$ sits at a uniformly random *nonzero* residue, safe with probability $(q-3)/(q-1)$, not $(q-2)/q$. Writing D for the divisor set,

$$\frac{\mathbb{E}[\text{twin} \mid F]}{p_0} = \prod_{q \in D} \underbrace{\frac{1}{(q-2)/q}}_{q/(q-2)} \cdot \prod_{q \notin D} \underbrace{\frac{(q-3)/(q-1)}{(q-2)/q}}_{g(q)}, \quad g(q) := \frac{q(q-3)}{(q-1)(q-2)} < 1.$$

Let $\mathcal{C} := \prod_{q>3} g(q)$. Since $\prod_{q \notin D} g(q) = \mathcal{C} / \prod_{q \in D} g(q)$ and $\frac{q/(q-2)}{g(q)} = \frac{q-1}{q-3}$, the divergent-looking expression collapses to a finite per-centre law:

$$\boxed{\frac{\mathbb{E}[\text{twin} \mid F]}{p_0} = \mathcal{C} \prod_{q|N} \frac{q-1}{q-3}}, \quad \mathcal{C} = \prod_{q>3} \frac{q(q-3)}{(q-1)(q-2)} = 0.72160. \quad (1)$$

Each factor of $\mathcal{C}$ is $1 - \frac{2}{(q-1)(q-2)} = 1 - O(1/q^2)$, so the product converges absolutely—the same $O(1/q^2)$ floor that makes the lattice-correlation products of Part XXII converge. The constant $\mathcal{C}$ is a genuine global suppression carried by the primes that miss N , and it is what the bare product $\prod q/(q-2)$ omits.

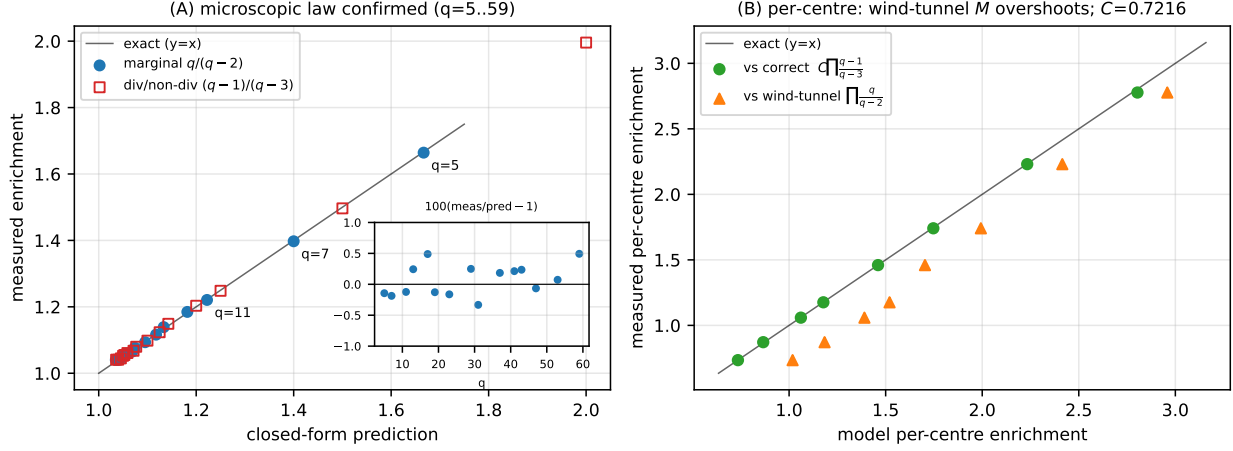


Figure 1: Auditing the closed form against real centres. (A) The microscopic marginal law: measured enrichment versus prediction, $q = 5, \dots, 59$, lies on $y = x$ for both the baseline-relative $q/(q-2)$ and the divisor-relative $(q-1)/(q-3)$ forms; residuals (inset) are $\pm 0.5\%$. (B) Per-centre enrichment versus model: the correct law $\mathcal{C} \prod (q-1)/(q-3)$ (green) sits on the diagonal, while the wind-tunnel object $\prod q/(q-2)$ (orange) systematically over-predicts the per-centre rate.

Equation (1) is not a fit: it is an exact CRT consequence, and it reproduces the data (Fig. 1B). Binning centres by their per-centre model value $\mathcal{C} \prod_{q|N} (q-1)/(q-3)$, the measured per-centre enrichment lies on the diagonal to within 1% across the entire range, while the wind-tunnel object $\prod q/(q-2)$ falls systematically below the diagonal—over-predicting the per-centre rate by a factor $\mathcal{C} \prod_{q|N} g(q)^{-1}$ that runs from $\mathcal{C} = 0.72$ for factor-poor centres up to 0.94 for small-prime-rich ones, exactly as (1) requires.

6 The $\times 7.03$ verdict

Two facts now sit cleanly side by side.

It is correct for the primorial centre. For the squarefree primorial Ω_{15} (exactly the factors $\{5, 7, \dots, 59\}$, nothing else), (1) gives $\mathcal{C} \prod_{q \leq 59} (q-1)/(q-3) = 6.983$, against the wind-tunnel $\prod_{q \leq 59} q/(q-2) = 7.030$: a gap of only -0.66% (Fig. 2B). The two nearly coincide because of an exact identity—the per-centre value equals $[\prod_{q \leq 59} q/(q-2)] \cdot \prod_{q > 59} g(q)$, and the tail $\prod_{q > 59} g(q) \approx 0.9934$ almost exactly cancels the difference between the $(q-1)/(q-3)$ boost on the fifteen small primes and the suppression $\mathcal{C}$. For the primorial centre itself, then, $\times 7.03$ is right to better than a percent. It is equally right as the *marginal* statement “centres divisible by $\{5, \dots, 59\}$ are $7.03\times$ more likely to be twin centres,” which is the multiplicative product confirmed in §4.

But it is an upper envelope, not the typical enrichment. The primorial centre is vanishingly rare. A *typical* centre at a given ω does not carry the smallest primes; it carries a mixture, mostly larger ones whose modifiers are close to 1. The measured average enrichment $\langle E \rangle(\omega)$ therefore sits far below the primorial envelope $\prod_{i \leq \omega} q_i/(q_i-2)$ at matched ω (Fig. 2A): at $\omega = 6$ the data give 2.63 against the envelope’s 4.27. The correct model average $\langle \mathcal{C} \prod (q-1)/(q-3) \rangle$ tracks the data

ω	n centres	twins	meas. $\langle E \rangle$	model (1)	primorial $\prod \frac{q}{q-2}$
1	6,741,131	120,175	0.731	0.722	1.667
2	14,241,165	311,377	0.896	0.891	2.333
3	9,700,927	276,430	1.168	1.171	2.852
4	2,449,886	94,528	1.581	1.614	3.370
5	197,049	10,654	2.216	2.297	3.820
6	2,958	190	2.632	3.313	4.269

Table 3: ω -stratified enrichment. The measured average tracks the per-centre model (1), both far below the primorial envelope at matched ω . The last row is sample-starved.

The $\times 7.03$ figure: correct as a marginal / primorial number, but an upper envelope vs typical ω

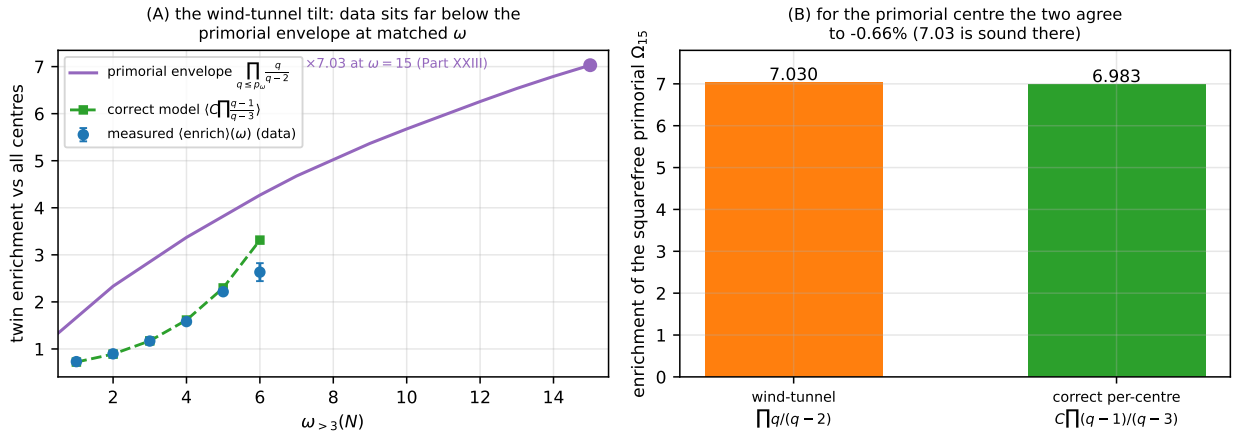


Figure 2: The $\times 7.03$ figure in context. (A) Measured average enrichment $\langle E \rangle(\omega)$ (blue, with Poisson bars) and the correct per-centre model (green) sit far below the primorial envelope (purple), which reaches $\times 7.03$ only at the $\omega = 15$ primorial. (B) For the squarefree primorial centre itself, the wind-tunnel product (7.030) and the correct per-centre value (6.983) agree to 0.66%.

through $\omega = 5$ (Table 3); the $\omega = 6$ bin, with 190 twins, is sample-starved (the -20% there is $\sim 3\sigma$ and consistent with either noise or a faint extra suppression). The headline $\times 7.03$ is thus the value reached only by the primorial-loaded subset of measure tending to zero, not the enrichment of the generic $\omega = 15$ centre.

7 Scope and honest ledger

The audit returns a clear verdict on a past extrapolation. The wind-tunnel *mechanism* is vindicated: the per-prime modifier $q/(q-2)$ is exact to $\lesssim 0.3\%$, and its multiplicativity holds within Poisson error throughout the reachable range. The headline number $\times 7.03$ is sound for what it actually describes—the squarefree primorial centre (to 0.66%) and the marginal divisibility statement (exactly, under multiplicativity). The audit adds two clarifications and no new theorem. First, an *interpretive* one: $q/(q-2)$ is a marginal factor; the per-centre likelihood of a centre with a given factorisation is $\mathcal{C} \prod (q-1)/(q-3)$ with $\mathcal{C} = 0.7216$, and using the bare product as a per-centre multiplier overpredicts non-primorial factorisations by up to $\sim 38\%$. This is also the clean reconciliation of the Part I $(q-1)/(q-3)$ form with the Part XXIII $q/(q-2)$ form, now seen as the per-centre and marginal

views of the identical two-killer geometry, joined by $g(q)$ and the constant $\mathcal{C}$. Second, a *scope* one: $\times 7.03$ is an upper envelope, realised by a vanishing subset, not the typical $\omega = 15$ enrichment.

We state the limits plainly. The data reach $6N \leq 2 \times 10^8$, smaller than the 6×10^8 run of Part XXVI [8] but ample for per-prime and per-centre statistics through $\omega \approx 6$; the behaviour at $\omega = 15$ remains an extrapolation, now built on ingredients measured to hold exactly where we can see them. The faint suppression lean in Tables 2 and 3 is the right sign for the Part IV second-order residual and the near-Poisson tilt of Part XXVI but does not exceed $\sim 2\text{--}3\sigma$; isolating it cleanly is a second-moment question we leave where Part XXVI left it. Everything here equals the discrete sieve count, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of twins or of any constellation. It is an audit of our own self-portrait, and the portrait holds.

Data and code availability. The sieve experiment (to $6N = 2 \times 10^8$), the per-prime, multiplicativity, per-centre and ω -stratified computations, the two figures, and a verifier of the per-centre identity (1) and the constant $\mathcal{C}$ are openly available at github.com/Ruqing1963/6N-windtunnel-audit.

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